

A van der Veen Hydrofracture Criterion as a Damage-Evolution Law

Technical note

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Abstract

The van der Veen crevasse models give a fracture-propagation or crack-arrest criterion, not a rate law. Here the fractured fraction of an ice column is written as $D = D_s + D_b$, where D_s and D_b are the fractions penetrated by surface and basal crevasses. The criterion is converted into an irreversible overstress law. The central result is the explicit source term

$$\mathcal{D}_t D = \langle 1 - D \rangle_+^p \left[\frac{1}{\tau_s} \langle \Psi_s \rangle_+^{m_s} + \frac{1}{\tau_b} \langle \Psi_b \rangle_+^{m_b} \right],$$

with either stress-intensity-factor drivers (the literal van der Veen LEFM model) or inexpensive crack-tip-stress drivers (the common reduced crevasse-depth model). The new timescales and rate exponents are constitutive parameters; they are not determined by the original static criterion.

1 State variables and conventions

Let $H > 0$ be the local ice thickness, d_s the downward penetration of surface crevasses, and d_b the upward penetration of basal crevasses. Define

$$D_s = \frac{d_s}{H}, \quad D_b = \frac{d_b}{H}, \quad D = D_s + D_b, \quad 0 \leq D \leq 1. \quad (1)$$

Thus D is a geometric, column-integrated damage variable: it is the fraction of the thickness intersected by surface or basal fractures. The constraint $D \leq 1$ assumes that the two fractured intervals do not overlap. Full-thickness connection, and hence the local calving/failure condition, is $D = 1$.

The material derivative following horizontal ice motion $\mathbf{u}$ is

$$\mathcal{D}_t(\cdot) = \frac{\partial(\cdot)}{\partial t} + \mathbf{u} \cdot \nabla(\cdot), \quad (2)$$

and $\langle x \rangle_+ = \max(x, 0)$. Damage is irreversible here; healing can be added as a separate, explicitly justified process if desired.

For a vertical fracture whose normal is the horizontal direction n , let R_n denote the tensile resistive stress. In a three-dimensional shallow stress model it should include the crevasse-parallel deviatoric stress; for example, $R_n = 2\tau_{nn} + \tau_{tt}$, rather than using τ_{nn} alone (Reynolds et al., 2025). Tension is positive.

2 The original criteria

2.1 Literal van der Veen LEFM criterion

The surface and bottom-crevasse models of [van der Veen \(1998b\)](#) and [van der Veen \(1998a\)](#) use linear elastic fracture mechanics (LEFM). For either crack $j \in \{s, b\}$, propagation occurs when

$$K_j > K_{Ic}, \quad (3)$$

and an advancing crack arrests where $K_j = K_{Ic}$. Here K_j is the net mode-I stress intensity factor and K_{Ic} is the fracture toughness of ice.

For completeness, write $a_s = HD_s$, let y measure depth downward from the upper surface, and let δ_s be the dry length above the water surface in a surface crack. A convenient statement of the van der Veen surface SIF is

$$K_s = F(\lambda_s)R_n\sqrt{\pi a_s} - \frac{2\rho_i g}{\sqrt{\pi a_s}} \int_0^{a_s} y G\left(\lambda_s, \frac{y}{a_s}\right) dy + \frac{2\rho_m g}{\sqrt{\pi a_s}} \int_0^{a_s} \langle y - \delta_s \rangle_+ G\left(\lambda_s, \frac{y}{a_s}\right) dy, \quad \lambda_s = \frac{a_s}{H}. \quad (4)$$

The three terms are the contributions of resistive stress, ice overburden, and surface meltwater. The water-column height is $h_s = \langle a_s - \delta_s \rangle_+$, and ρ_m is the meltwater density.

For a basal crack, let y measure upward from the bed, let p_b^0 be basal water pressure at the crack mouth, and let ρ_b be the water density within the crack. The corresponding weight-function form is

$$K_b = \frac{2}{\sqrt{\pi a_b}} \int_0^{a_b} \left[R_n - \rho_i g(H - y) + \langle p_b^0 - \rho_b g y \rangle_+ \right] G\left(\lambda_b, \frac{y}{a_b}\right) dy, \quad \lambda_b = \frac{a_b}{H}. \quad (5)$$

The finite-slab functions F and G are listed in [Appendix A](#). Equations (4) and (5) make all hydrostatic assumptions explicit and are directly usable in a quadrature routine.

LEFM requires an existing flaw. Numerically, initialize $D_j \geq D_{j0} > 0$, or evaluate K_j using $a_j = H \max(D_j, D_{j0})$. The seed size D_{j0} is a model parameter, not a prediction of LEFM.

2.2 Reduced crack-tip-stress criterion

A widely used, cheaper closure places the crack tip where the net opening stress vanishes. It is often grouped with “van der Veen-type” or crevasse-depth hydrofracture laws, but it is not the same approximation as the LEFM criterion above. Define

$$\Sigma = \frac{R_n}{\rho_i g H}, \quad W_s = \frac{h_s}{H}, \quad r_s = \frac{\rho_m}{\rho_i}, \quad r_b = \frac{\rho_b}{\rho_i}, \quad (6)$$

and the basal height above water-pressure support

$$H_{ab} = H - \frac{p_b^0}{\rho_i g}, \quad A_b = \frac{H_{ab}}{H}. \quad (7)$$

For marine-connected basal water, $p_b^0 = \rho_b g w$, where w is water depth at the bed. The reduced basal expression below assumes that water reaches the basal crack tip. The dimensionless crack-tip opening stresses are then

$$\Psi_s^{ZS} = \Sigma + r_s W_s - D_s, \quad (8)$$

$$\Psi_b^{ZS} = \Sigma - A_b - (r_b - 1) D_b. \quad (9)$$

The label ZS denotes “zero stress.” Positive values open the crack; zero values give the usual static depths

$$D_s^* = \Sigma + r_s W_s, \quad (10)$$

$$D_b^* = \frac{1}{r_b - 1} \langle \Sigma - A_b \rangle_+ = \frac{\rho_i}{\rho_b - \rho_i} \left\langle \frac{R_n}{\rho_i g H} - \frac{H_{ab}}{H} \right\rangle_+. \quad (11)$$

Equation (11) assumes $\rho_b > \rho_i$. These are the familiar surface and basal crevasse-depth formulas used in large-scale calving parameterizations (Nick et al., 2010; Reynolds et al., 2025).

3 Damage-evolution law

3.1 Generic kinetic wrapper

Define a dimensionless fracture driver for each crack family:

$$\Psi_j^{\text{LEFM}} = \frac{K_j}{K_{\text{Ic}}} - 1 \quad (\text{literal van der Veen}), \quad \text{or} \quad \Psi_j = \Psi_j^{\text{ZS}} \quad (\text{reduced model}). \quad (12)$$

The proposed evolution equations are

$$\mathcal{D}_t D_s = \frac{\langle 1 - D \rangle_+^p}{\tau_s} \langle \Psi_s \rangle_+^{m_s}, \quad (13)$$

$$\mathcal{D}_t D_b = \frac{\langle 1 - D \rangle_+^p}{\tau_b} \langle \Psi_b \rangle_+^{m_b}, \quad (14)$$

$$\mathcal{D}_t D = \langle 1 - D \rangle_+^p \left[\frac{1}{\tau_s} \langle \Psi_s \rangle_+^{m_s} + \frac{1}{\tau_b} \langle \Psi_b \rangle_+^{m_b} \right]. \quad (15)$$

Here $\tau_j > 0$ is a propagation/relaxation timescale, $m_j \geq 1$ is a rate-sensitivity exponent, and $p > 0$ controls smooth saturation as the remaining ligament vanishes. The minimal choice is

$$m_s = m_b = p = 1. \quad (16)$$

Equation (15) is the requested right-hand side for the damage material derivative. It has four useful properties:

1. it is dimensionally correct because each Ψ_j is dimensionless and τ_j has units of time;
2. the positive part enforces irreversibility and reproduces the original propagation threshold;
3. static crack-arrest states are unchanged: growth stops when the driver first reaches zero;
4. the saturation factor makes the interval $0 \leq D \leq 1$ invariant for the continuous equations.

The original van der Veen work does not determine τ_j , m_j , or p . They must be calibrated, resolved as fast brittle timescales, or taken to an instantaneous limit. For $p > 0$, $D = 1$ is approached asymptotically, so a numerical model should declare calving at $D \geq 1 - \epsilon_{\text{calv}}$. If finite-time arrival at exactly $D = 1$ is required, omit the saturation factor for $D < 1$ and project the update onto the admissible set after each step.

3.2 Fully explicit reduced law

Substituting equations (8) and (9) gives an algebraic right-hand side with no SIF quadrature:

$$\mathcal{D}_t D = \langle 1 - D \rangle_+^p \left[\frac{1}{\tau_s} \langle \Sigma + r_s W_s - D_s \rangle_+^{m_s} + \frac{1}{\tau_b} \langle \Sigma - A_b - (r_b - 1) D_b \rangle_+^{m_b} \right]. \quad (17)$$

This is the most convenient form for a depth-integrated ice-flow model. The two component equations, rather than the total equation alone, must be advanced because the right-hand side depends separately on D_s and D_b .

For constant water-column height, the surface term relaxes toward equation (10). Hydrology can instead make W_s a function of damage. In particular, a fully water-filled surface crack has $W_s = D_s$, so

$$\Psi_s^{ZS} = \Sigma + (r_s - 1) D_s. \quad (18)$$

Because freshwater is denser than ice, $r_s > 1$; crack deepening then increases the driver and produces the expected hydrofracture runaway. A water-volume or water-supply equation is therefore needed if h_s is not prescribed.

4 Coupling and implementation choices

Stress–damage feedback. The R_n in the driver should be the trial resistive stress consistent with the ice-flow solver. If a fixed membrane force is carried only by the intact ligament, a simple load-sharing closure is

$$R_n^{\text{lig}} = \frac{N_n}{H(1 - D)}, \quad (19)$$

which adds destabilizing feedback. Do not apply equation (19) a second time if the constitutive or flow solver already accounts for the loss of load-bearing area.

Thickness evolution. Equations (13)–(15) evolve the fraction D_j as the material state. If a code instead evolves dimensional crack depths, the exact kinematic conversion is

$$\mathcal{D}_t d_j = H \mathcal{D}_t D_j + D_j \mathcal{D}_t H. \quad (20)$$

One should decide whether changing thickness stretches the interpreted crack depth or whether D_j is retained as a history variable; mixing the two interpretations introduces a spurious damage source or sink.

Time stepping. At each material point: (i) compute trial stress and water state; (ii) evaluate the two drivers; (iii) advance both component equations; (iv) enforce non-negativity and cap $D_s + D_b \leq 1$; and (v) apply the calving threshold. Explicit stepping should resolve the smaller of τ_s and τ_b , or use substepping/implicit integration.

Geometry warning. The van der Veen weight functions describe a finite rectangular slab with a single edge crack. They are not universal weight functions; grounded basal boundary conditions can materially change the SIF, especially for deep cracks (Jiménez and Duddu, 2018). For a grounded-glacier application, retain the kinetic wrapper in equations (13)–(15) but replace K_j by one computed with a geometry-appropriate weight function or numerical fracture solution.

5 Recommended minimal model

For a first implementation in a large-scale model, use equation (17) with $m_s = m_b = p = 1$, advect D_s and D_b separately, and calve at $D \geq 1 - \epsilon_{\text{calv}}$. This gives the compact system

$$\begin{aligned}\mathcal{D}_t D_s &= \frac{1-D}{\tau_s} \langle \Sigma + r_s W_s - D_s \rangle_+, \\ \mathcal{D}_t D_b &= \frac{1-D}{\tau_b} \langle \Sigma - A_b - (r_b - 1) D_b \rangle_+, \quad D = D_s + D_b.\end{aligned}\tag{21}$$

When literal adherence to van der Veen LEFM is required, keep exactly the same evolution equations and replace the two bracketed drivers with $\langle K_j/K_{\text{Ic}} - 1 \rangle_+$, using equations (4) and (5).

A van der Veen finite-slab weight functions

The functions reproduced by Jiménez and Duddu (2018) are

$$F(\lambda) = 1.12 - 0.23\lambda + 10.55\lambda^2 - 21.72\lambda^3 + 30.39\lambda^4,\tag{22}$$

and

$$\begin{aligned}G(\lambda, \gamma) &= \frac{3.52(1-\gamma)}{(1-\lambda)^{3/2}} - \frac{4.35 - 5.28\gamma}{(1-\lambda)^{1/2}} \\ &\quad + \left[\frac{1.30 - 0.30\gamma^{3/2}}{(1-\gamma^2)^{1/2}} + 0.83 - 1.76\gamma \right] [1 - (1-\gamma)\lambda],\end{aligned}\tag{23}$$

where $\lambda = a/H$ and $\gamma = y/a$. The crack-tip singularity in G is integrable. In computation, use endpoint-aware quadrature and stop the LEFM calculation short of $\lambda = 1$, where the slab formula is singular and the column has already met the calving criterion.

References

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